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Infections and antibiotic treatment in long-term care facilities: results from 1-year cross sectional study in three Polish settings

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Abstract

Background Health care-associated infections (HAIs) in long-term care facilities (LTCFs) represent a significant issue within the health care system. With the expected increase in the geriatric population, they will pose an increasing challenge to the health care system. The aim of this study was to analyse and compare the epidemiology of HAIs and the use of antimicrobial agents in Polish LTCFs participating in a longitudinal study based on the European Centre for Disease Control and Prevention (ECDC).

Materials and methods This study utilized the H4LS: Healthcare-Associated Infections in Long-Term Care Facilities—Longitudinal Study protocol from ECDC. Additionally, data on antibiotic consumption, reflecting the distribution according to specific groups and the WHO AWaRe classification system, were collected. The study was conducted in nursing homes. Patient observation took place from February 1, 2022, to January 31, 2023.

Results This study included 269 patients residing in three nursing homes. The average age of the residents was 84.06 years (SD 10.2). To varying degrees, cognitive impairments were present in 86.99% of the residents; mobility limitations in 89.96%; urinary or faecal incontinence in 92.19%; urinary catheters in 14.5%; and vascular catheters in 6.69%. Comparison between settings revealed significant differences only for Charlson comorbidity index and urinary/ fecal incontinence. A total of 420 infections were recorded among 183 residents. The cumulative incidence rate was 5.2 per 1000 patient-days. The most prevalent infections were pneumonia (35.24%) and urinary tract infections (20.95%). For pneumonia and conjunctivitis there were also observed differences in incidence between study settings. Among the HAIs, 382 (90.95%) were treated with antimicrobial agents. The most commonly used antibiotics were third-generation cephalosporins (20.83%) and penicillin combinations, including beta-lactamase inhibitors (17.80%) and fluoroquinolones (13.83%). However, significant differences were observed for antibiotic treatment patterns in study settings. In stratification according to the WHO AWaRe classification, the most commonly used antimicrobials belonged to the Watch group (49%).

Conclusions These results highlight the urgent need for the effective implementation of infection prevention and control in LTCFs. The implementation should be performed at the regional or country level because only multicentre

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studies allow us to obtain the most reliable data. Our results revealed a lack of antimicrobial stewardship in the study of LTCFs. This situation also demands quick interventions.

Keywords Long-term care, Nursing homes, Infection control, Antibiotics use

Background

Health care-associated infections (HAIs) in long-term care facilities (LTCFs) represent a significant issue within the health care system, lowering the quality of life for affected individuals and potentially leading to premature deaths [1]. According to data from the European Centre for Disease Prevention and Control (ECDC), every day, 1 in 26 residents of European LTCFs suffers from an HAI, and more than half of these infections could be prevented [1]. Data presented by the CDC indicate that in the United States, this issue affects 1 in 43 LTCF patients [2]. The problem of HAIs in LTCFs is not extensively covered in the available literature [3]. The primary data on this topic come from small local studies and the periodic, Europe-wide HALT point prevalence surveys organized by ECDC [4–6]. Therefore, the scale of the problem is likely underestimated. With the expected increase in the geriatric population, HAIs in LTCFs are expected to pose an increasingly serious challenge to the health care system. Data presented by Statistics Poland (GUS) show that in 2019, Poles over the age of 65 constituted 18.1% of the general population, with this age group projected to account for 23.3% in 2030 and over 30% by 2050 [7]. In 2019, approximately 1.34% of the population over the age of 65 in Poland received assistance through long-term care services (both residential and nonresidential) [8]. According to the Polish Ministry of Health, in 2021, there were 461 residential long-term care facilities in Poland, with a total of approximately 27,000 beds, all of which were fully occupied [9]. The types of HAIs characteristic for LTCFs include respiratory tract infections, urinary tract infections, skin and soft tissue infections, and gastrointestinal infections, including *C. difficile* [10, 11]. In recent years, an alarming increase in the prevalence of colonization by multidrug-resistant microorganisms has been observed not only in Poland and Europe but also globally, affecting both the general population and residents of LTCFs [12–16]. These microorganisms, which are abundant in the LTCF environment due to factors such as shared sanitary facilities, vulnerability of elderly individuals based on general health and immune system and frequent interaction between healthcare providers and residents, can become aetiological agents of symptomatic infections [17]. Proper infection surveillance and prevention, as well as effective antimicrobial stewardship, are essential for reducing the frequency of HAIs and the occurrence of multidrug-resistant microorganisms among LTCF residents [18–21].

The aim of this study was to analyse and compare the epidemiology of HAIs and the use of antimicrobial

agents in three Polish LTCFs participating in a study based on the ECDC H4LS (Healthcare-Associated Infections in Long-Term Care Facilities—Longitudinal Study) protocol.

Material and methods

The study utilized the H4LS Study protocol from the ECDC. This protocol includes questionnaires designed to collect data on patients, the setting (LTCF), and infections (HAIs), as well as the criteria and definitions of infections and the structural indicators that determine infection-related morbidity, which are specific to this type of health care [22]. Additionally, beyond the scope of the data included in the H4LS protocol, the study collected information on antibiotic consumption, including the name of each antibiotic used and the indication (infection/prophylaxis). The distribution of antibiotics—the number of prescriptions per 100 patients, according to specific groups—was chosen as one of the outcomes. Additionally, we used the WHO AWaRe classification system to present the distribution of antibiotics from three groups—Access, Watch and Reserved. The AWaRe system is a compiled list of 180 commonly used antibiotics in three categories. The Access group consists of first-choice agents for common infections, generally with a narrower spectrum of action. The Watch group contains antibiotics that are more commonly used as second-line treatments or for treating resistant organisms. The Reserved group contains antibiotics considered drugs of last resort, when no other options are available [23].

The study was conducted in three long-term care facilities in Kraków, the capital of Małopolska Province. All the facilities were nursing homes, with medical personnel in place: nurses and physicians. Patient observation took place from February 1, 2022, to January 31, 2023. Data were collected directly from the residents and from their medical records prospectively and recorded weekly by infection control nurse trained in the protocol. Definitions of infection cases are presented in Supplementary file. The participation of the facilities in the study was voluntary.

Analyses were conducted in R version 4.5.1 (RStudio 2025.05.1). Differences in proportions between long-term care facilities (LTCFs) were evaluated using chi-squared tests and test for equality of proportions with continuity correction, including pairwise post hoc comparisons. Incidence rates adjusted for exposure time (patient-days) were analyzed using Poisson regression models with log link and offset. Incidence rate ratios

(IRRs) with 95% confidence intervals were derived using the emmeans package. The Kruskal–Wallis test was used for comparing non-normally distributed variables. Statistical significance was set at $\alpha = 0.05$.

Results

The study included 269 patients residing in three nursing homes. The average age of the residents was 84.06 years (SD = 10.2), with those over 85 years of age comprising 52.04% ($n = 140$) of the studied population. Most patients were women—78.07% ($n = 210$). Cognitive impairments of varying degrees were present in 86.99% ($n = 234$) of the residents; mobility limitations of varying degrees were observed in 89.96% ($n = 242$); urinary or faecal incontinence was reported in 92.19% ($n = 248$); urinary catheters were maintained in 14.5% ($n = 39$) of the residents; and vascular catheters were maintained in 6.69% ($n = 18$). The detailed patient characteristics, including the evaluated factors, are presented in Table 1. The median Charlson Comorbidity Index score for the entire study population was 4 points.

Statistically significant differences between facilities were found for the Charlson Comorbidity Index ($p = 0.006$) and for urinary/faecal incontinence ($p < 0.030$). Post hoc analysis showed a higher Charlson Index in LTCF 3 compared to LTCF 1 ($p = 0.008$). Although the global difference in incontinence rates was significant, none of the pairwise comparisons reached statistical

significance after correction; however, the comparison between LTCF 1 and LTCF 3 showed a borderline trend ($p = 0.054$). No significant differences were observed in mobility ($p = 0.109$).

A total of 420 infections were recorded among 183 residents, with 68% of the residents experiencing at least one infection. The most prevalent infections were pneumonia (35.24%), urinary tract infections (20.95%), and other respiratory tract infections (13.18%) (nonpneumonic, non-COVID-19, including pharyngitis, common cold, flu and lower respiratory tract infections)—Table 2. Most other respiratory tract infections can be assumed to be viral, as no laboratory, microbiological or radiological diagnostics were conducted in these cases to confirm otherwise. In one of the facilities (LTCF3), the incidence of skin infections reached 10.95%, and in one facility (LTCF2), the incidence of gastrointestinal infections was 11.58%. *Clostridioides difficile* infections accounted for 25.8% ($n = 8$) of all gastrointestinal infections. No cases of surgical site infections, sinusitis, or oral candidiasis were recorded (types specifically listed in the protocol)—Table 2.

The cumulative incidence rate, calculated per patient-day, was 5.2 per 1000 patient-days. The incidence of infections varied between long-term care facilities (LTCFs), both in terms of specific infection types and overall burden. Pairwise comparisons of infection episode rates, adjusted for patient-days, revealed significantly higher

Table 1 Characteristics of the study population – HAI risk factors according to facility

		n = 69		n = 89		n = 111				p
		LTCF 1	%	LTCF 2	%	LTCF 3	%	Total	%	
Patient characteristics	Average age	83.94		82.15		85.66		84.06		Kruskall-Wallis $p = 0.137$
	SD	9.11		21.10		9.05		10.2		
	Age > 85	35	50.72	42	47.19	63	56.76	140	52.04	Chisq $p = 0.391$
	Male gender	14	20.29	26	29.21	19	17.12	59	21.93	Chisq $p = 0.113$
Cognitive impairments	None	10	14.49	13	14.61	12	10.81	35	13.01	Chisq $p = 0.804$
	Mild	11	15.94	19	21.35	19	17.12	49	18.22	
	Moderate	24	34.78	24	26.97	32	28.83	80	29.74	
	Severe	24	34.78	31	34.83	46	41.44	101	37.55	
	Unknown	0	0.00	2	2.25	2	1.80	4	1.49	
Mobility	mobile/independent	9	13.04	13	14.61	5	4.50	27	10.04	Chisq $p = 0.109$
	Wheelchair-bound	30	43.48	41	46.07	46	41.44	117	43.49	
	Chronically bedridden	30	43.48	33	37.08	54	48.65	117	43.49	
	Unknown	0	0.00	2	2.25	6	5.41	8	2.97	
Urinary/faecal incontinence	Yes	59	85.50	82	92.13	107	96.40	248	92.19	*Chisq $p < 0.030$
	No	10	14.50	7	7.87	4	3.60	21	7.81	
Urinary catheter	Yes	11	15.94	15	16.85	13	11.71	39	14.50	Chisq $p = 0.546$
	No	58	84.06	74	83.15	98	88.29	230	85.50	
Vascular catheter	Yes	2	2.90	7	7.87	9	8.11	18	6.69	Chisq $p = 0.3425$
	No	67	97.10	82	92.13	102	91.89	251	93.31	
Charlson comorbidity index	Median	4		4		5		4		**Kruskal–Wallis $p = 0.006$
	IQR	3		4		4				

*For urinary/ fecal incontinence pairwise with Bonferroni corr. 1–2: $p = 0.848$; 1–3: $p = 0.054$; 2–3: $p = 0.949$. **For Charlson comorbidity Index Dunn after K–W 1–2: $p = 1.000$; 1–3: $p = 0.008$; 2–3: $p = 0.078$

Table 2 Distribution of infection types according to facility

Infection	LTCF 1	%	LTCF 2	%	LTCF 3	%	Total	%	LTCF1 / LTCF2 IRR (95% CI) p	LTCF1 / LTCF3 IRR (95% CI) p	LTCF2 / LTCF3 IRR (95% CI) p
UTI	34	36.56	29	15.26	25	18.25	88	20.95	1.54 (0.85–2.78) p=0.2054	2.12 (1.14–3.94), p=0.0119	1.38 (0.73–2.62) p=0.4631
Pneumonia	34	36.56	62	32.63	52	37.96	148	35.24	0.72 (0.44–1.19), p=0.269	1.02 (0.61–1.71), p=0.995	1.42 (0.91–2.21), p=0.149
Other respira- tory tract infections	9	9.68	31	16.32	16	11.68	56	13.18	0.38 (0.16–0.92), p=0.0289	0.88 (0.33–2.33), p=0.9475	2.31 (1.12–4.75), p=0.0182
COVID-19	2	2.15	11	5.79	4	2.92	17	4.00	0.24 (0.04–1.44), p=0.149	0.78 (0.1–5.94), p=0.956	3.27 (0.83–12.87), p=0.105
Skin infections	6	6.45	8	4.21	15	10.95	29	6.82	0.98 (0.28–3.49), p=0.999	0.62 (0.2–1.94), p=0.592	0.64 (0.23–1.77), p=0.553
Conjunctivitis	5	5.38	19	10.00	10	7.30	34	8.00	0.34 (0.11–1.12), p=0.0862	0.78 (0.22–2.82), p=0.8931	2.26 (0.91–5.65), p=0.0919
Gastrointestinal infections	2	2.15	22	11.58	7	5.1	31	7.29	0.12 (0.02–0.67), p=0.01108	0.45 (0.07–2.92), p=0.57217	3.74 (1.35–10.35), p=0.00666
Other	1	1.08	8	4.21	8	5.84	17	4	0.16 (0.01–1.97), p=0.203	0.2 (0.02–2.34), p=0.272	1.19 (0.37–3.84), p=0.935
Total	93		190		137		420		0.64 (0.48–0.86), p=0.00131	1.06 (0.77–1.45), p=0.90341	1.65 (1.27–2.15), p= <0.001
Patient-days (ptds)	20,689		27,116		32,287		80,092				
Incidence den- sity/1000 ptds	4.5		7.0		4.2		5.2				

incidence of total infections in LTCF2 compared to LTCF1 (IRR=0.64; 95% CI 0.48–0.86; p=0.00131). Conversely, LTCF2 also showed a significantly higher burden of gastrointestinal infections (IRR=3.74; 95% CI 1.35–10.35; p=0.007) and other respiratory infections (IRR=2.31; 95% CI 1.12–4.75; p=0.018) compared to LTCF3.

LTCF3 exhibited a significantly lower rate of urinary tract infections compared to LTCF1 (IRR=2.12; 95% CI: 1.14–3.94; p=0.004), and also showed a trend toward lower rates of conjunctivitis and COVID-19, although these differences did not reach statistical significance—Table 2.

Rates of pneumonia and skin infections did not significantly differ between facilities. While some pairwise differences in rarer infection types (e.g. “Other”) were numerically notable, they did not reach statistical significance, likely due to limited counts.

No statistically significant relationships were found between the presence of specific infection risk factors and the frequency of HAIs.

The incidence of infections was analysed across different months of the year (Fig. 1). Statistically significant seasonality (meteorological seasons) was observed for certain infections. Pneumonia occurred more frequently in winter than in spring (p=0.012) and in summer (p=0.019); other respiratory tract infections were more common in winter than in summer (0.015); skin and soft tissue infections were more common in spring

and winter than in summer (p=0.016, p=0.048); and *C. difficile* infections were more common in spring than in autumn (p=0.035).

During the year under study, 79 of the participants (29.37% of the observed population) passed away (Table 3). Among the recorded deaths, 49.37% (n=39) were directly or indirectly related to an infection, resulting in a HAI-related fatality rate of 9.28%. Statistical analysis did not reveal any significant differences (p=0.2244) between facilities in terms of HAI-related mortality (cases with HAI as the direct or indirect cause).

A total of 180 patients (66.91%) received at least one antimicrobial agent during the observation.

Among the HAIs, 382 (90.95%) were treated with antimicrobial agents (Fig. 2). In 75.13% (n=287) of the therapies, a single antimicrobial agent was used; in 14.92% (n=57) of the cases, two agents were used; in 6.54% (n=25) of the cases, three were used; in 3.14% (n=12) of the cases, four were used; and one infection (0.26%) was treated with five antimicrobial agents. These drugs were used either in combination or in sequence.

All patients with pneumonia and urinary tract infections were treated with antibiotics. Most (73.21%, n=41) cases of respiratory infections other than pneumonia (classified as colds, flu, pharyngitis, and lower respiratory tract infections) were also treated with antibiotics, despite the likely viral aetiology. Additionally, 64.71% (n=11) of COVID-19 cases were treated with antibiotics regardless of clinical severity—94.12% (n=16) of the

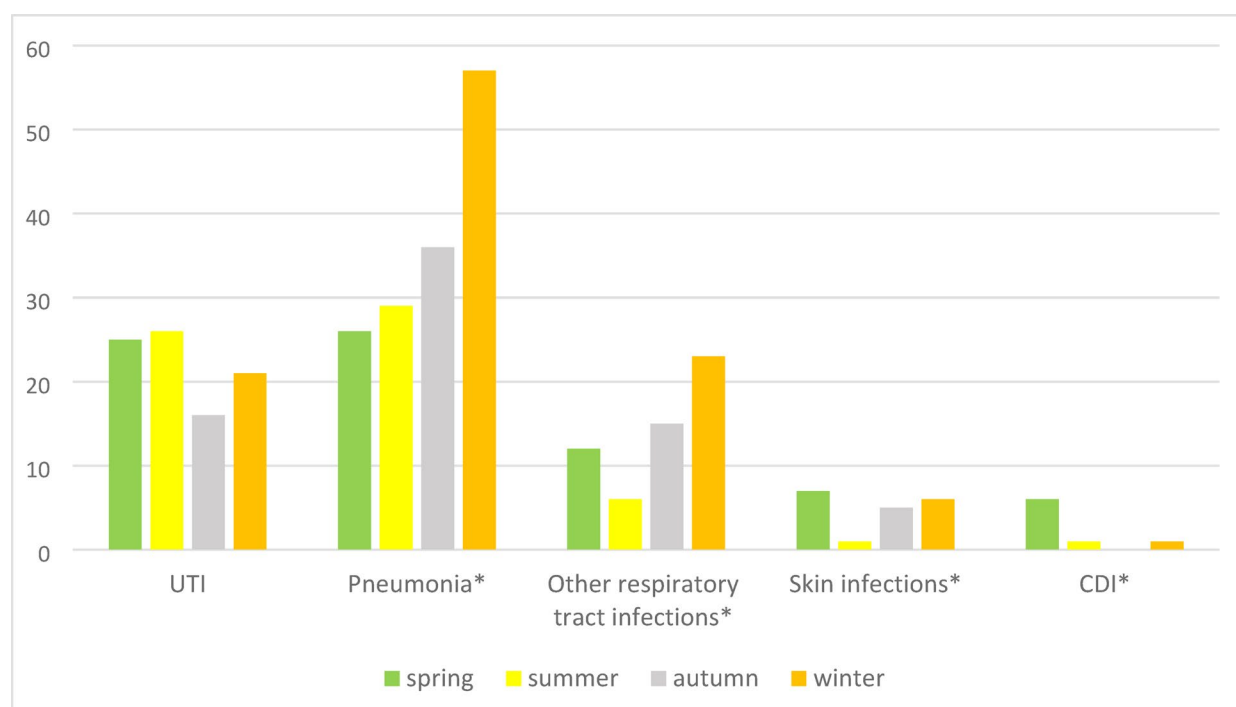


Fig. 1 Seasonality of infections—statistically significant differences indicated with asterisks (*)

Table 3 Patients mortality over one year of observation

Deaths	LTCF 1	%	LTCF 2	%	LTCF 3	%	Total	%
Deaths—total	23		22		34		79	
Deaths caused by HAIs	13	56.52	13	59.09	13	38.24	39	49.37
HAI as the sole (direct) cause	11	84.62	7	53.85	12	92.31	30	76.92
HAI as the indirect cause	2	15.38	6	46.15	1	7.69	9	23.08

recorded COVID-19 cases were classified as asymptomatic or mild/moderate, with only 5.88% ($n=1$) classified as severe. Detailed breakdown of antimicrobial prescriptions by the type of infection treated is presented in Fig. 3.

The group of antibiotics most commonly administered was third-generation cephalosporins, which accounted for 20.83% of all prescribed antimicrobial agents; penicillin combinations, including beta-lactamase inhibitors, accounting for 17.80%; and fluoroquinolones, accounting for 13.83% (Table 4).

No statistically significant difference was found between the facilities regarding the overall percentage of infections treated with antibiotics. However, statistically significant differences were observed between facilities with respect to the use of the two dominant groups of antibiotics (i.e., third-generation cephalosporins and penicillin with beta-lactamase inhibitors), but not for fluoroquinolones. Third-generation cephalosporins were used significantly more frequently in LTCF1, whereas penicillins with beta-lactamase inhibitors were more commonly used in LTCF3. Detailed data are presented in Table 4.

Urinary tract infections were most commonly treated with fluoroquinolones ($n=23$, 26.14%), whereas pneumonia was primarily treated with third-generation cephalosporins and penicillin with a beta-lactamase inhibitor: 49.32% and 43.24%, respectively.

Vancomycin was used most frequently in the treatment of gastrointestinal infections (75%), half of which were classified as CDI; vancomycin was also used in the treatment of pelvic abscess and pyometra.

No antibiotics for prophylaxis were registered

In stratification according to the WHO AWaRe classification, the most commonly used antimicrobials belonged to the Watch group (49%), and no antimicrobial agents from the Reserved group were used (Fig. 4).

Microbiological tests were conducted for 10.95% of the infections (only for UTIs and diarrhoea), ranging from 9.68% in LTCF1 to 12.6% in LTCF3.

Serological tests and molecular techniques were applied only for diagnosing COVID-19 (4.05% of all HAIs).

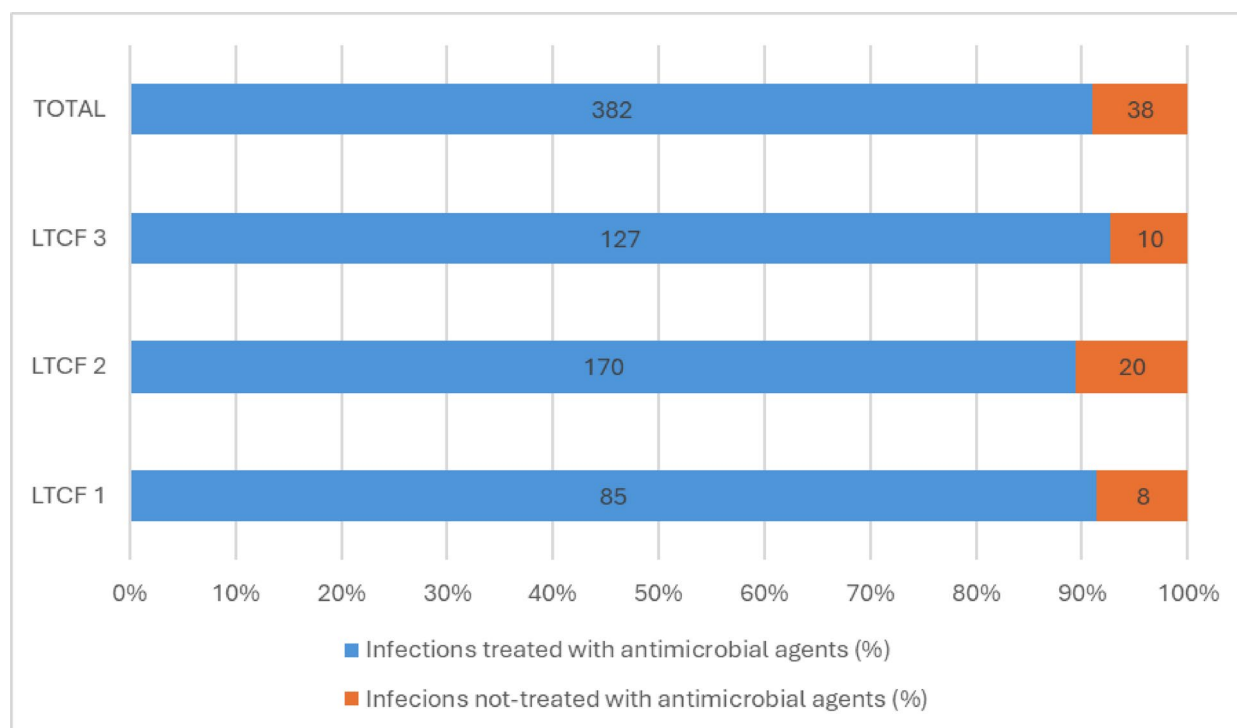


Fig. 2 Percentage of infections treated by antimicrobial agent

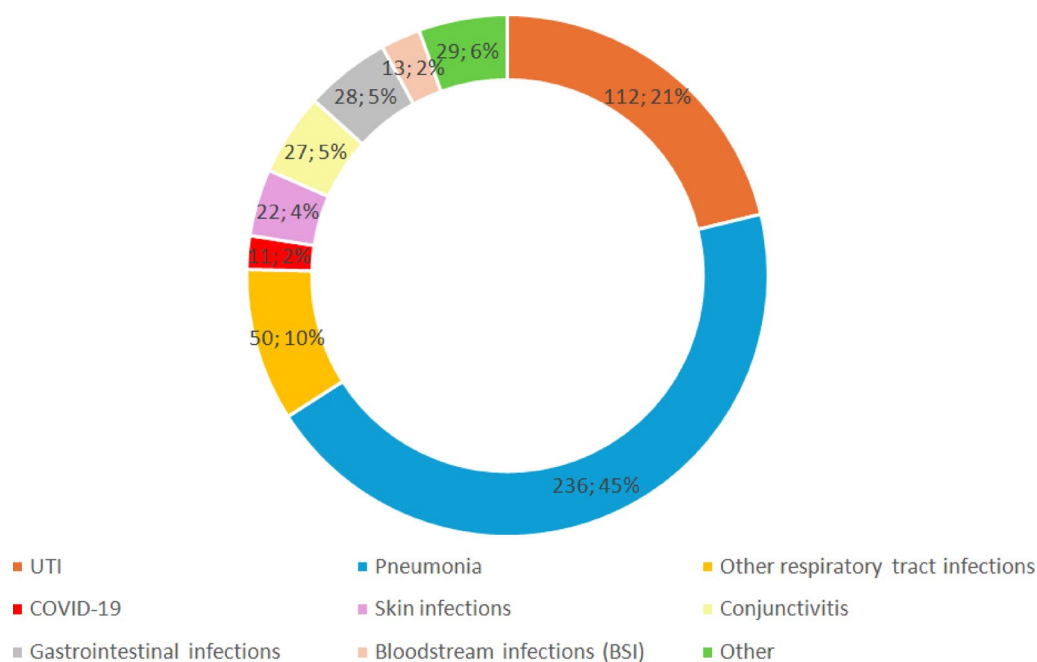


Fig. 3 Breakdown of antimicrobial prescriptions by the type of infection treated

Discussion

The HAI incidence rate observed in the study was 5 per 1000 patient-days, which aligns with expectations. Several studies on the occurrence of HAIs in European LTCFs have reported incidence rates ranging from 2.7 to 11.8 per 1000 patient-days [10, 24–28].

The most common infections were pneumonia, urinary tract infections, and other (nonpneumonic, non-COVID) respiratory infections, which correspond to the frequency of infection types reported in similar studies and in the general geriatric population [10, 11, 25, 29].

Table 4 Consumption of specific antibiotic groups according to facility

Antimicrobial agent	LTCF 1		LTCF 2		LTCF 3		Total	%	p	Proportion test		
	n	%	n	%	n	%				LTCF 1/LTCF 2	LTCF 1/LTCF 3	LTCF 2/LTCF 3
Fluoroquinolones	11	9.57	41	16.47	21	12.80	73	13.83	0.187	na	na	na
Penicillins	2	1.74	2	0.80	1	0.61	5	0.95	0.6	na	na	na
Penicillin + Inhibitor	25	21.74	24	9.64	45	27.44	94	17.8	<0.001	0.00289	0.347	<0.001
2nd Generation Cephalosporins	0	0.00	18	7.23	4	2.44	22	4.17	0.00239	0.00699	0.24	0.0578
3rd Generation Cephalosporins	41	35.65	42	16.87	27	16.46	110	20.83	<0.001	<0.001	<0.001	1
Metronidazole	4	3.48	25	10.04	6	3.66	35	6.63	0.0119	0.0522	1	0.0266
Vancomycin	0	0.00	7	2.81	1	0.61	8	1.52	0.0648	na	na	na
Nifuroxazide*	1	0.87	9	3.61	0	0.00	10	1.89	0.0204	0.252	0.858	0.0342
Macrolides	4	3.48	24	9.64	9	5.49	37	7.01	0.0664	na	na	na
Aminoglycosides	18	15.65	15	6.02	8	4.88	41	7.77	0.00154	0.00547	0.00454	0.781
Lincosamides	1	0.87	3	1.20	2	1.22	6	1.14	0.954	na	na	na
Furagin	3	2.61	13	5.22	7	4.27	23	4.36	0.524	na	na	na
Fosfomycin	0	0.00	1	0.40	6	3.66	7	1.33	0.00676	1	0.0981	0.0341
Tetracyclines	1	0.87	1	0.40	0	0.00	2	0.38	0.506	na	na	Na
Others	2	1.74	16	6.43	14	8.54	32	6.06	0.0609	na	na	Na
Azoles	2	1.74	1	0.40	4	2.44	7	1.33	0.189	na	na	na
Trimethoprim + Sulfamethoxazole	0	0.00	7	2.81	9	5.49	16	3.03	0.0301	0.16	0.0271	0.263
Total	115		249		164		528		ns			

na Not applicable, ns Not significant.

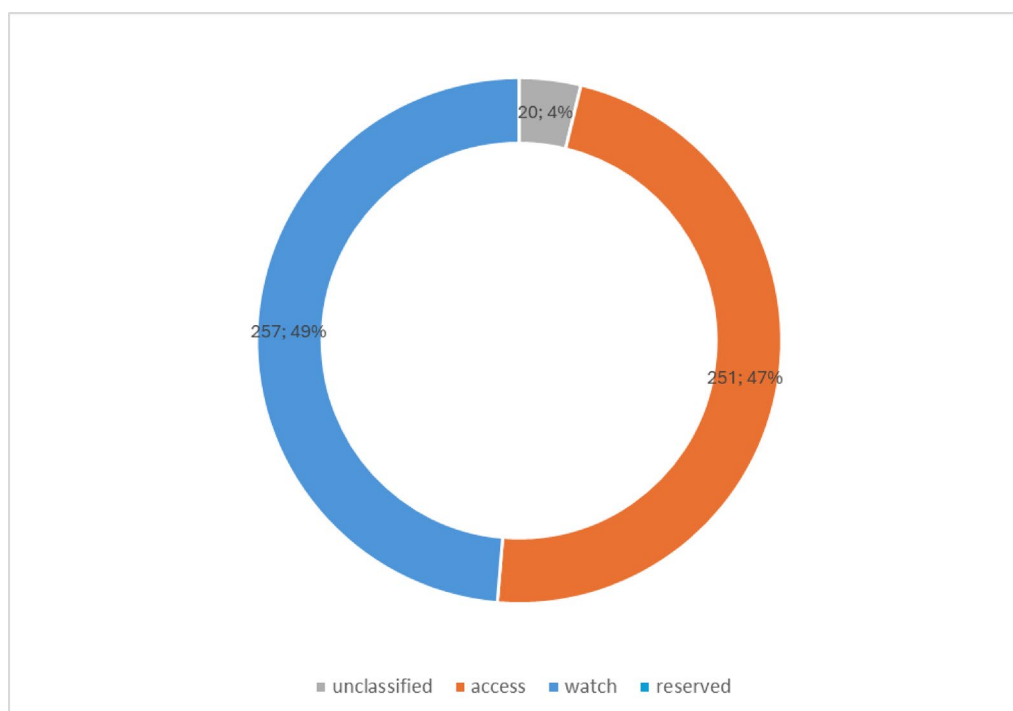


Fig. 4 Usage of antimicrobial agents divided by WHO AWaRe classification

However, our studied population, compared with residents of European long-term care facilities, had a lower average age but a greater burden of risk factors for HAIs [10]. All our study settings were nursing homes and in Poland they are basically dedicated to persons with quite severe health conditions. Contrary European data come from different type of LTCFs (most commonly nursing and residential homes) what may be the reason. There were no statistically significant differences between facilities in terms of risk factor burden, except for the Charlson Comorbidity Index and urinary/ faecal incontinence. Estimated 10-year survival for patient 80 years or older with Charlson Comorbidity Index of 4 is 53% and with Index of 5–21% [30] which indicates high importance for prognosis of LTCF3 patients compared to LTCF1 for which such a statistically significant difference was observed. This would indicate significant impact on prevalence and mortality of HAIs. However, our data do not show significant differences between LTCFs with respect to deaths, as well as no statistically significant relationships were found between the presence of specific infection risk factors and the frequency of HAIs. Percentage of infections treated with antibiotic did not differ significantly between facilities. Such observation most likely is due to the study size, especially regarding analyses according to the type of infection.

Nevertheless, our results confirm some expected facts. The observed seasonality of respiratory tract infections and skin and soft tissue infections in relation to

meteorological seasons aligns with common trends observed in the general population [31–33]. Additionally, a statistically significant increase in the incidence of CDI occurs in the months following an elevated period of respiratory infections treated with antibiotics, which is a well-known risk factor for the development of *C. difficile* infection [34–36]. This observation, although expected and convergent with common trends, has important implications for infection surveillance and antimicrobial stewardship. Given the limited use of microbiological testing (only 10.95% on average), the treatment is often empirical. Hamard et al. also found that antimicrobial prescribing in French nursing homes was mainly empirical [37] due to lack of access and delay in obtaining diagnostic tests, but also equipment, time, skills and habits of the nursing Staff. Thus, there is a field for improvement, namely, the implementation of microbiological diagnostics, which could help increase the percentage of targeted antimicrobial therapy. However, quick increase of number of infection cases with microbiological tests are difficult to achieve. Thus, the improvement could include, as proposed by participants of the study of Hamard, access to guidelines tailored to NH's context and decision support tools, developing dialog with an infectious disease specialist by building a partnership with a hospital-based infectious disease department, audit and feedback interventions, training for all members of NH's Staff on different subjects related to AMS [37].

Additionally, our observation of the seasonality of infections indicates the need for the implementation of prospective, continuous infection surveillance in long-term care facilities. The epidemiological data on infections in such settings in European countries and in Poland are gathered mainly in point-prevalence studies. As a result, they may be either under- or overestimated. The study of Różańska et al. of surgical site infections in hospitals revealed, the variability of infections may be due not only to meteorological seasons but also to organizational factors [38]. Thus, the effective implementation of prospective and continuous infection surveillance, considering antimicrobial stewardship, among other factors, is extremely important. For now, in Poland, we face a lack of information, which was observed in our study. Antibiotic therapy is widely used for the treatment of infections, particularly urinary tract infections, pneumonia, and other (nonpneumonic) respiratory infections, but most of this treatment is empirical therapy. COVID-19 infections are also predominantly treated with antibiotics, regardless of their clinical severity. In light of the current consensus on the aetiology and treatment of respiratory infections, it can be concluded that antibiotic use is likely excessive, especially in the treatment of respiratory infections [39, 40]. This microbiological confirmation of only 10.95% of infections arouses special concern because it is almost two times lower than that reported in the Polish LTCFs HALT-3 study [41]. The facilities covered by our study were nursing homes, and all of them had physicians employed in the facility. In the HALT-3 project, half of the facilities were residential homes with no medical personnel in place. This observation confirms the urgent necessity of the introduction of effective infection control programs in long-term care, which should be followed by increasing patient safety and decreasing the numbers of infections and related deaths. In our study, HAIs were directly or indirectly associated with 49.37% of the deaths recorded during the observation period. Although the mortality rate due to infections was lower than that reported in European studies [20], it still represents a high proportion of potentially avoidable deaths.

In our study, the most commonly used antibiotic groups were third-generation cephalosporins, penicillin combinations including beta-lactamase inhibitors, and fluoroquinolones. Previous studies conducted in Polish LTCFs reported greater variability in the types of antibiotics used [42]. The widespread use of the aforementioned antibiotics is concerning, particularly with respect to fluoroquinolones, which carry a high risk of CDI and complications related to the nervous, vascular, and musculoskeletal systems and are no longer recommended for empirical therapy [43]. The low usage of penicillin is also concerning, as it is widely recommended as a first-line drug in the treatment of pneumonia and other

respiratory tract infections [38]. Our analysis, which was based on the WHO AWaRe system, also revealed no optimal antimicrobial treatment; 49% of all antibiotics were from the Watch group, and 47% were from the Access group. For comparison, in the Belgian long-term care population over 65 years of age, Access group antibiotics were used in 70.4%, and Watch antibiotics were used in 24.6% and 23.9% [44]. The WHO goal for Access group is at least 70% of global human antibiotics use [45] and Belgian data confirm possibility of achieving this goal.

Limitations of the study

Although the study presents the results from one year of observation and is not a single-centre study, the small number (three) of settings is the first limitation. Additionally, all the settings were nursing homes, which is advantageous for providing a detailed comparison of infection epidemiology and antibiotic consumption but is limited by the wider possible context. From point prevalence studies covering different types of settings, i.e., nursing and residential homes, we know that there may be important differences. An additional limitation in light of the proper qualification of infections and validation of antibiotic treatment patterns is the very small percentage of cases with microbiological confirmation of infection aetiology.

Conclusions

These results highlight the urgent need for the implementation of an infection prevention and control system based on continuous surveillance. Infection control based on point prevalence may not capture all important data for prevention strategies.

Additionally, infection prevention and control systems should be implemented at the regional or country level because only multicentre studies can obtain the most reliable data.

Our results revealed a lack of implementation of antimicrobial stewardship in the study of LTCFs, including insufficient performance of microbiological tests, which is likely a general case in Poland. This situation also demands quick interventions, such as staff training, use of rapid testing and developing national guidelines.

Abbreviations

HAI	Healthcare-associated infection
LTCF	Long term care facility
HALS	Healthcare-associated infections in long-term care facilities longitudinal study
ECDC	The European Centre for Disease Control and Prevention
UTI	Urinary tract infection
CDI	Clostridioides difficile infection
SD	Standard deviation

Supplementary Information

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Supplementary Material 1.

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Author contributions

Conceptualization KB-T, AR; Methodology, AR, KL, ER; Formal Analysis, KB-T, KS; Investigation KB-T, AT; Data Curation KS, KB-T, Writing—Original Draft Preparation, KB-T Writing—Review and Editing AR, KL, ER, JW-M.

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Declarations

Ethics approval and consent to participate

Informed consent for patient participation in the study was waived by the Bioethics Committee of Jagiellonian University in Krakow, which approved the study under approval number KBET (1072.6120.247.2022).

Competing interests

The authors declare no competing interests.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

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